Business Intelligence BSIT 701

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Python 3.14 + Jupyter Notebook

Group 3		
Student Number	**Initials**	**Surname**
26091590	S.G	Lunga
21742106	S.S	Shezi
22315958	A.E	Zulu
22326212	S.N	Jele
21814236	S.S	Ndlela
22334812	A	Sosibo
22256578	S.P	Buthelezi
22228521	S.M	Mthiyane
22227345	M.S	Sthole
22258409	N.A	Simamane

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CHURN ANALYSIS ASSIGNMENT REPORT

1. INTRODUCTION

This work is about predicting which customers might leave, estimating future sales, and figuring out what steps to take. We're using Python and machine learning to do this. The main aim is to give business leaders a clearer picture of how customers behave.

We want to help them spot customers who are likely to churn, forecast how much money each customer could bring in, and then suggest smart actions to keep those customers and boost profits. We're working with data from an e-commerce company about customer behavior.

This data includes details about the customers themselves, what they buy, how well their orders are delivered, what they do on the site, and how satisfied they are.

2. OBJECTIVES

The main objectives of this project were:

- Try to figure out which customers were likely to stop using our service.
- Estimate how much money we expected to get from our customers.
- Find out the biggest reasons why customers either leave or spend more with us.
- Come up with practical advice for the business, all based on what our analysis showed.

3. TOOLS AND TECHNOLOGIES USED

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost
- SHAP
- Power BI (for data understanding)

4. DATA PREPARATION

The dataset was loaded into the Jupyter Notebook using pandas. Initial exploration showed 5000 rows and multiple customer-related features.

Data cleaning and feature engineering included:

- Handling missing values
- Creating Recency, Frequency, and Monetary (RFM) features
- Creating Total_Spend
- Creating Discount Rate %
- Creating customer priority groups
- Creating churn target variable

Final customer-level dataset size:
5000 rows × 18+ engineered features

5. TASK 1: CUSTOMER CHURN PREDICTION

5.1 Target Variable

The target variable used for churn prediction was:

Churn
1 = Customer likely to churn
0 = Customer retained

Class distribution after SMOTE balancing:

Churn 1 = 2548
Churn 0 = 2548

5.2 Models Used For this part, we trained these classification models:

• Logistic Regression

• Random Forest Classifier

• XGBoost Classifier

5.3 Model Performance Here's how the models performed:

Logistic Regression's accuracy: 0.9774

Random Forest's accuracy: 1.0000

XGBoost's accuracy: 1.0000

5.4 Interpretation

The Random Forest and XGBoost models did an excellent job predicting things, much better than the Logistic Regression model did. This tells us that how customers churn in our data is quite predictable, especially when we use these more advanced machine learning methods. We turned to SHAP analysis to figure out why the models made their choices and to really nail down what makes customers leave.

The main things that push customers away turned out to be:

• Monetary

• Average Order Value

• Frequency

• Average Delivery Days

• Age These points really play a big role in whether a customer sticks around or decides to walk away.

6. TASK 2: REVENUE FORECASTING

6.1 Target Variable

For predicting how much money we'll make, we decided to look at:

Total_Spend

This amount really gives us the best picture of a customer's value over time and how much they add to our total income..

6.2 Models Used

The following regression models were trained:

• Linear Regression
• Random Forest Regressor

6.3 Model Performance

Linear Regression:

MAE: 1315.75
RMSE: 2520.67
R^2 Score: 0.089

Random Forest Regressor:

MAE: 1413.11
RMSE: 2554.22
R^2 Score: 0.065

6.4 Interpretation

Linear Regression outperformed Random Forest Regressor by achieving lower error values and a higher R^2 score.

This suggests that customer revenue behavior in this dataset follows a more linear pattern than expected, making Linear Regression the more suitable model.

Top revenue-driving features included:

• Average Discount
• Age
• Average Delivery Days
• Average Pages Viewed
• Average Session Duration

These variables provide strong insight into customer spending behavior.

7. TASK 3: PRESCRIPTIVE ANALYTICS

7.1 Customer Segmentation

Customers were segmented into three priority groups using monetary value:

• Low Value
• Medium Value
• High Value

Distribution:

Low Value = 1667
Medium Value = 1666
High Value = 1667

7.2 Churn vs Customer Priority

Analysis showed:

• Low-value customers had the highest churn
• High-value customers showed the strongest retention
• Medium-value customers showed balanced churn behavior

This supports targeted business strategy development.

7.3 Business Recommendations

Strategy 1: Protect High-Value Customers

• VIP loyalty programs
• Premium customer support

• Personalized offers
• Faster delivery priority

Strategy 2: Improve Medium-Value Customers

• Upselling and cross-selling
• Personalized engagement campaigns
• Product recommendations

Strategy 3: Reduce Low-Value Customer Churn

• Re-engagement campaigns
• Discount optimization
• First-purchase incentives

Strategy 4: Optimize Discount Policies

• Avoid excessive discounting
• Use targeted promotions
• Behavior-based discount strategies

Strategy 5: Improve Delivery Performance

• Reduce delivery delays
• Improve logistics efficiency
• Strengthen delivery communication

8. CONCLUSION

This project did a good job using predictive and prescriptive analytics, along with machine learning, to solve some real business problems. We used customer churn prediction to find out which customers might leave.

Revenue forecasting helped us guess how much customers were worth. And prescriptive analytics turned what we learned from the models into actual decisions for the business. When we looked at classification models, Random Forest and XGBoost worked best for guessing who would leave, but for figuring out future revenue, Linear Regression did a better job than Random Forest Regressor.

The suggestions we came up with help management keep more customers, make more money, and make things run smoother. This project really shows how useful data analytics can be when making important business choices.

9. REFERENCE

Dataset

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10. APPENDIX

The full Python implementation for data preprocessing, feature engineering, model training, and evaluation is provided in the attached Jupyter Notebook and PDF document.

The notebook includes:

- Data loading and cleaning
- Feature engineering (RFM metrics, churn variable)
- Machine learning models:
    - Logistic Regression
    - Random Forest
    - XGBoost
- Model evaluation metrics
- Revenue forecasting models
- SHAP explainability

All intermediate exploratory steps (such as data previews and checks) were excluded from the PDF version to ensure clarity and readability.